Class Exercise.

Consider a sinusodial transverse wave travelling in the positive x-direction. This wave has a wavelength of 20 cm & a period of 10 s. The amplitude of this wave is 4 cm.

i) Determine the necessary conditions for this wave to be plotted as a positive or negative sine function.

ii) Find the wave function : $y(x,t)$.

iii) Calculate $y(x=0, t=5s)$

iv) Determine its longitudinal velocity at $x=0$, $t=5s$.